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Effect of laser energy density on microstructures and mechanical properties of selective laser melted Ti-6Al-4V alloy

Ti-6Al-4V samples were fabricated by selective laser melting (SLM) with a series of different laser energy inputs. The microstructure presented a feature of fine acicular martensite due to the high cooling rate of the SLM process. All the relative density, micro-hardness and tensile properties initially increased and then decreased with increasing laser energy density. The fracture morphology exhibited a mixed mode of complex ductile and brittle fracture. All the results indicated that the laser energy density has a crucial effect on the microstructures and mechanical properties of the SLMed Ti-6Al-4V alloy. The correlations between the laser energy density and the microstructures and mechanical properties of the SLMed Ti-6Al-4V alloy were studied.

Keywords: Selective laser melting; Ti-6Al-4V alloy; Laser energy density; Microstructures; Mechanical properties

1. Introduction

Three-dimensional (3D) printing, also known as rapid prototyping or additive manufacturing, is a technology used to directly manufacture 3D objects layer by layer from digital models [1–3]. Selective laser melting (SLM) is a 3D printing technique that uses a laser beam as the energy source to rapidly melt metal powders. The molten metal pool then rapidly cools down, and the consolidated material starts to form the product [4–6]. SLM offers several advantages such as low material waste, short design and production cycle, ability to handle complex shapes and intrinsic engineered features compared to conventional technologies such as casting and forging [7, 8].

Ti-6Al-4V alloy is extensively used in the aerospace, automobile, marine, chemical and biomedicine fields because of its low density, high specific strength, superior corrosion resistance, excellent biocompatibility and mechanical properties [9–11]. In the last decade, much research has been conducted to develop and improve the SLM of Ti-6Al-4V alloy [12–22]. Generally, the mechanical properties of SLMed samples are highly influenced by the average applied laser energy per unit volume of material, described by laser energy density (*LED*) as follows [23, 24]:

$$LED = \frac{P}{vhd} \quad (1)$$

Where *P*, *v*, *h* and *d* are laser power, scanning speed, hatch spacing and layer thickness, respectively. A pulse laser as energy source applied in SLM, *v* can be expressed as

$$v = \frac{P_d}{T_e} \quad (2)$$

where *P_d* and *T_e* are point distance and exposure time, respectively.

Han et al. [25] studied the mechanical properties including the relative density, microhardness and tensile strength of Ti-6Al-4V samples fabricated by SLM using a constant laser power (194 W) and layer thickness (20 μm) while the scanning speed ranged from 400 to 1100 mm · s⁻¹ and the hatch spacing ranged from 70 to 120 μm. Kasperovich et al. [26] prepared numbers of SLMed Ti-6Al-4V samples to obtain a minimum porosity volume fraction. The results indicated that insufficient or excessive *LED* led to binding faults and round pores, while intermediate *LED* exhibited the lowest porosity volume fraction in the SLMed samples. Bartolomeu et al. [27] reported that there was an optimized processing window where *LED* ranged from 41 to 99 J · mm⁻³ to manufacture Ti-6Al-4V samples with high microhardness and strength. However, a better understanding is needed of the influence of *LED* on the microstructure and mechanical properties of SLMed Ti-6Al-4V; especially, the inherent mechanism of *LED* should be investigated further. In addition, very few works have studied the optimized microstructure and mechanical properties as a function of *LED* of Ti-6Al-4V samples fabricated by SLM with a pulsed laser as energy source.

In this study, by taking the energy input as the main factor, we aimed at studying the influence of *LED* by varying two important processing parameters, laser power and point distance, on the microstructures and mechanical properties (the relative density, micro-hardness, tensile properties and fracture mechanism) of the SLMed Ti-6Al-4V samples, and obtaining a reliable *LED* process window for the SLMed Ti-6Al-4V samples.

2. Experimental procedure

Ti-6Al-4V alloy spherical powders were produced by vacuum induction gas atomization (with a particle size of 10–50 μm, supplied by Zhonghang Maite Co. Ltd., China). The morphologies of the Ti-6Al-4V spherical powders

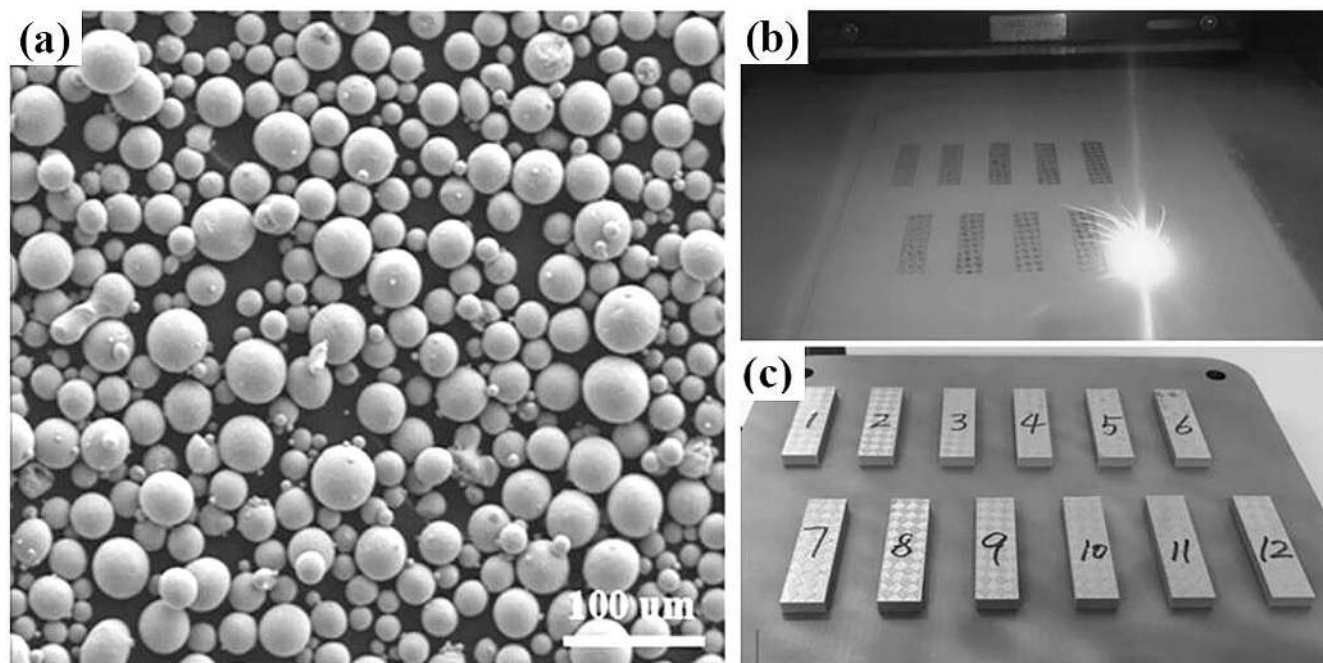


Fig. 1. (a) Morphologies of the spherical Ti-6Al-4V powders, (b) the working process of the SLM, and (c) the SLMed samples.

were examined using a field emission scanning electron microscope (FESEM, JEOL, JSM-7800F, Japan) as shown in Fig. 1a. The chemical compositions of the Ti-6Al-4V powders are listed in Table 1.

The spherical powders were processed by a selective laser melting machine (AM250, Renishaw Co., Ltd., UK) that employed a pulsed fiber laser with a wavelength of 1070 nm with a maximum average power of 200 W. During the SLM process, the chamber was flooded with argon until a value of below 1000 ppm oxygen content was reached. The working process of SLM and the produced samples are shown in Fig. 1b and c, respectively. In this study, the laser spot diameter, hatch spacing, layer thickness and exposure time were 70, 125, 50 μm and 125 μs , respectively. The process parameters and the calculated LED of the SLMed Ti-6Al-4V samples are listed in Table 2.

The SLMed samples were mechanically ground with silicon-carbide abrasive paper, and fine surfaces were obtained by polishing the samples with velvet and diamond polishing solution. The surface morphologies of the polished but unetched samples were observed via an optical microscope (OM, Zeiss Axio Scope A1, Germany). The microstructures of the SLMed samples were examined using an OM and an SEM, before this analysis, the polished surfaces of the samples were etched by a solution of HNO_3 in ethanol with a volume ratio of 5:95.

The density was measured using an electronic densimeter (BR-120, Bairong, China) which was based on the Archimedeian method. Micro-hardness testing was performed

under a 100 g load and a 10 s dwell time using a Micro Vickers hardness tester (Ahvd-1000XYZ, China). Tensile testing was conducted at room temperature using a universal testing machine (MTS TestStar 810, USA) with a tensile speed of $0.2 \text{ mm} \cdot \text{min}^{-1}$. The fracture morphology was also examined in the SEM.

3. Results and discussion

Figure 2a and b shows the microstructures of a typical SLMed Ti-6Al-4V sample with an LED of $66.7 \text{ J} \cdot \text{mm}^{-3}$. It can be seen that fine acicular martensitic microstructure was formed due to the very fast cooling rate during the SLM process [28, 29]. This is in good agreement with the microstructure obtained by Agius et al. [16] with an LED of $68.37 \text{ J} \cdot \text{mm}^{-3}$. Figure 2c–e shows the surface morphologies of some SLMed Ti-6Al-4V samples. The SLMed Ti-6Al-4V sample with a lower LED of $40.0 \text{ J} \cdot \text{mm}^{-3}$ exhibited a large number of pores, especially distributed along the laser scanning tracks, owing to incomplete melting of the Ti-6Al-4V powders. The sample SLMed at medium

Table 2. Process parameters and LED of the SLMed Ti-6Al-4V samples.

Laser power (W)	Point distance (μm)	Laser energy density ($\text{J} \cdot \text{mm}^{-3}$)
180	90	40.0
200	90	44.4
180	75	48.0
200	75	53.3
180	60	60.0
200	60	66.7
180	45	80.0
200	45	88.9

Table 1. Chemical composition and oxygen content of the spherical Ti-6Al-4V powders used in the experiment (wt.%).

Element	Al	V	Fe	C	O	Ti
Content	5.8	4.1	0.3	0.1	0.13	Balance

LED of $66.7 \text{ J} \cdot \text{mm}^{-3}$ exhibited an almost pore-free morphology, only very small pores with a diameter of several microns were sporadically observed. At a higher LED of $88.9 \text{ J} \cdot \text{mm}^{-3}$, some defects formed due to excessive energy input into the Ti-6Al-4V alloys.

Figure 3 presents the relative density as a function of the LED of the SLMed Ti-6Al-4V samples. The value of the relative density increased with increasing LED before it

reached a maximum value and then decreased with further increasing LED . The maximum value of the relative density, 99.2%, was achieved at the LED of $66.7 \text{ J} \cdot \text{mm}^{-3}$. This means that the sample was well densified and did not have obvious pores in the microstructure.

At lower LED , the Ti-6Al-4V power could not be completely melted, and caused higher viscosity and poor wettability of Ti-6Al-4V in the melt pool [30, 31], the liquid

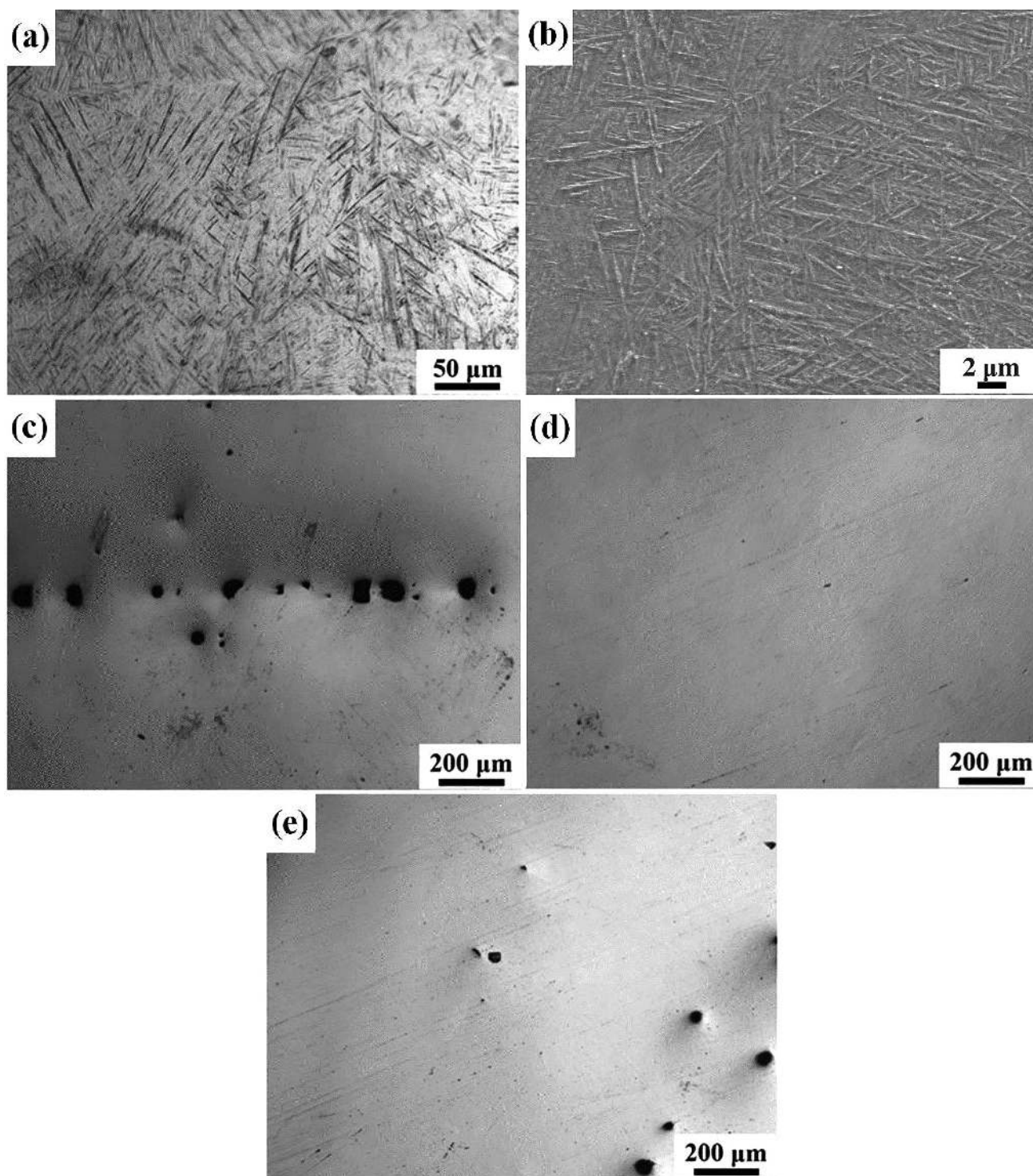


Fig. 2. (a) Optical metallographic view (polished and etched), and (b) SEM image (polished and etched) of the SLMed Ti-6Al-4V alloy with an LED of $66.7 \text{ J} \cdot \text{mm}^{-3}$. Morphologies (polished and unetched) of the SLMed Ti-6Al-4V samples with an LED of (c) $40.0 \text{ J} \cdot \text{mm}^{-3}$, (d) $66.7 \text{ J} \cdot \text{mm}^{-3}$, and (e) $88.9 \text{ J} \cdot \text{mm}^{-3}$.

phase of the Ti-6Al-4V alloy was difficult to spread between two adjacent laser scanning tracks, thereby, the width and depth of the melt pool were smaller than the hatch spacing and layer thickness, respectively. Some pores inevitably appeared between adjacent laser scanning tracks, thus leading to lower relative density of the Ti-6Al-4V alloy.

With increasing *LED*, the viscosity of the liquid phase of the Ti-6Al-4V in the melt pool decreased and the flowability increased. Hence, the diffusion of the liquid phase resulted in an increase in the dimensions of the melt pool. The width and depth of the melt pool were slightly greater than the hatch spacing and layer thickness, respectively, leading to excellent metallurgical bonding between two adjacent laser scanning tracks and the number of pores decreased. Hence, the optimal value of the relative density of the Ti-6Al-4V alloy was achieved.

With further increasing *LED*, the laser irradiation cause not only a larger melt pool but also extra heat that may evaporate the low melting point elements in the liquid phase of the Ti-6Al-4V alloy. In addition, high purity argon was used as circulation gas in the SLM process. Due to fast solidification during the SLM process, the trapped gases inside the melt pool formed pores [26], thereby decreasing the relative density of the Ti-6Al-4V alloy. Furthermore, a large temperature gradient comes into play at high *LED*, resulting in higher surface tension gradient and stronger Marangoni convection [32]. Hence, over-melting and over-heating of liquid phase led to degraded stability and thus the relative density of the Ti-6Al-4V alloy decreased.

Figure 4 shows the effect of the *LED* on the micro-hardness and ultimate tensile strength of the SLMed Ti-6Al-4V samples. The micro-hardness and ultimate tensile strength initially increased as the *LED* increased. Both the micro-hardness and ultimate tensile strength reached a maximum value at an *LED* of $66.7 \text{ J} \cdot \text{mm}^{-3}$, they were 408.9 HV and 1380 MPa, respectively. Then they decreased as the *LED* further increased. Figure 5 illustrates the stress-strain curves of some typical SLMed Ti-6Al-4V samples with different *LED*s. The tensile properties are summarized in Table 3. Although the SLMed Ti-6Al-4V samples exhibited relatively low ductility due to quite fine needle-like marten-

sitic microstructure, the tensile yield strength, ultimate tensile strength, strain to fracture and elongation had maximum values at an *LED* of $66.7 \text{ J} \cdot \text{mm}^{-3}$. By considering the relative density, microhardness and tensile properties, one estimates that the optimized *LED* should be $66.7 \text{ J} \cdot \text{mm}^{-3}$ for the SLMed Ti-6Al-4V samples. This is in good agreement with the optimized *LED* reported by Bartolomeu et al. [27].

At insufficient *LED*, the Ti-6Al-4V powders were only partially melted because of the laser's energy shortage, the melt pool was small, so that the contact area between the melt pool and solidified powders was limited, resulting in low wettability, worsened flowing, unfavorable spreading and corresponding balling effect [33]. The balling effect may result in the formation of discontinuous scan tracks and hinder uniform deposition of a fresh powder layer on the previously processed layer. Thus, the poor metallurgical bonding of adjacent scanning tracks led to poor mechanical properties include microhardness, tensile properties and fracture behavior of the Ti-6Al-4V alloy.

At excessive *LED*, over-melting and over-heating of melt pool led to higher instability and even splashing of the liq-

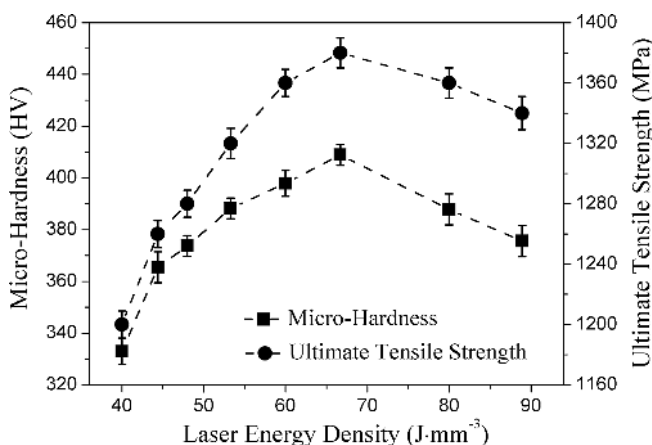


Fig. 4. Microhardness and ultimate tensile strength as functions of the *LED* of the SLMed Ti-6Al-4V samples.

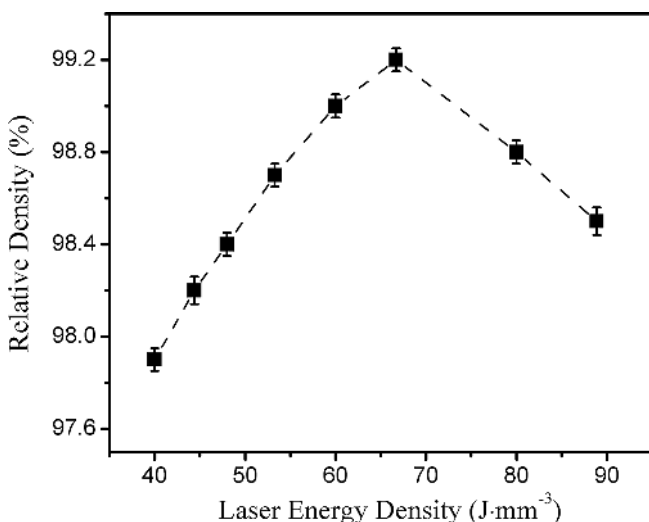


Fig. 3. Relative density as a function of the *LED* of the SLMed Ti-6Al-4V samples.

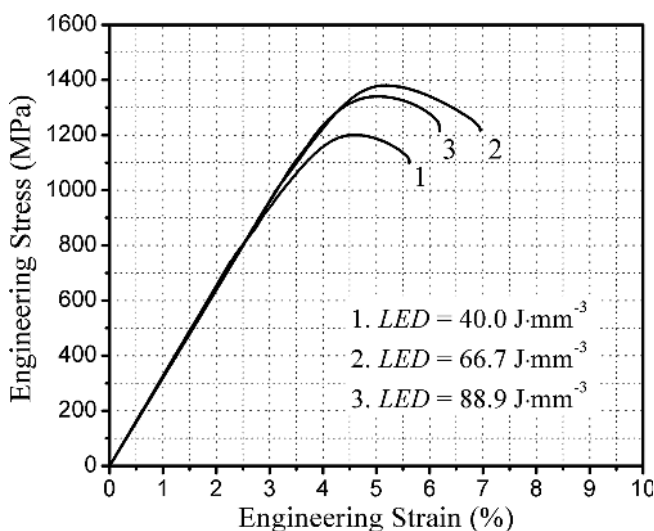


Fig. 5. Stress-strain curves of the SLMed Ti-6Al-4V samples.

Table 3. Tensile properties of the SLMed Ti-6Al-4V samples.

Laser energy density ($\text{J} \cdot \text{mm}^{-3}$)	Tensile yield stress (at 0.2% strain) (MPa)	Ultimate tensile strength (MPa)	Strain to fracture (%)	Elongation (%)
40.0	1021 ± 9	1200 ± 6	5.62 ± 0.1	2.23 ± 0.2
66.7	1273 ± 9	1380 ± 6	6.95 ± 0.1	3.18 ± 0.2
88.9	1249 ± 9	1337 ± 6	6.19 ± 0.1	2.40 ± 0.2

uid phase. In addition, the melted Ti-6Al-4V powder grains stuck together and formed a series of spheres due to excessive shrinkage and high residual stresses under the action of higher surface tension. Furthermore, the low melting point elements in the Ti-6Al-4V alloy probably evaporated due to overhigh LED, the evaporated metal gases and inert gas (argon) trapped within the melt pool may form pores during fast solidification of the SLM process, as reported by Kasperovich et al. [26] and Vilaro et al. [34]. Generally, the balling effect, gas entrapments and defective pores may result in the poor metallurgical bonding. More details can be found in Refs. [35–42]. Thus, the mechanical properties of the Ti-6Al-4V alloy degraded.

Figure 6 shows the tensile fracture morphologies of a typical Ti-6Al-4V sample with an LED of $66.7 \text{ J} \cdot \text{mm}^{-3}$. It

exhibited features of ductile–brittle hybrid fracture, which predominantly showed plenty of small and shallow ductile-dimples with diameters of 2–6 μm , however, some quasi-cleavage fracture patterns observed in some local zones, indicating the relatively low ductility of the SLMed Ti-6Al-4V sample [43]. The result is in good agreement with its stress–strain curve. This kind of fracture mode is attributed to the high brittleness and low ductility of the quite fine acicular martensitic microstructure formed due to the very fast cooling rate during the SLM process.

4. Conclusions

Ti-6Al-4V samples were fabricated by selective laser melting. The microstructure and mechanical properties of the SLMed Ti-6Al-4V alloys were studied. The following conclusions can be drawn:

1. The microstructure of the SLMed Ti-6Al-4V samples consisted mainly of fine acicular martensite. The SLMed Ti-6Al-4V sample at medium LED of $66.7 \text{ J} \cdot \text{mm}^{-3}$ exhibited an almost pore-free morphology.
2. With increasing LED, the relative density of the SLMed Ti-6Al-4V sample initially increased and then decreased. The maximum relative density of 99.2% occurred at an LED of $66.7 \text{ J} \cdot \text{mm}^{-3}$.
3. The microhardness and tensile properties of the SLMed Ti-6Al-4V samples first increased and then decreased with the increase in LED. The maximum value of the microhardness, tensile yield strength, ultimate tensile strength, strain to fracture and elongation was 408.9 HV, 1273 MPa, 1380 MPa, 6.95% and 3.18%, respectively, at an LED of $66.7 \text{ J} \cdot \text{mm}^{-3}$.
4. The tensile fracture of the SLMed Ti-6Al-4V sample exhibited a feature of ductile–brittle hybrid fracture, which predominantly showed plenty of small and shallow ductile-dimples together with some quasi-cleavage fracture patterns in some local zones.

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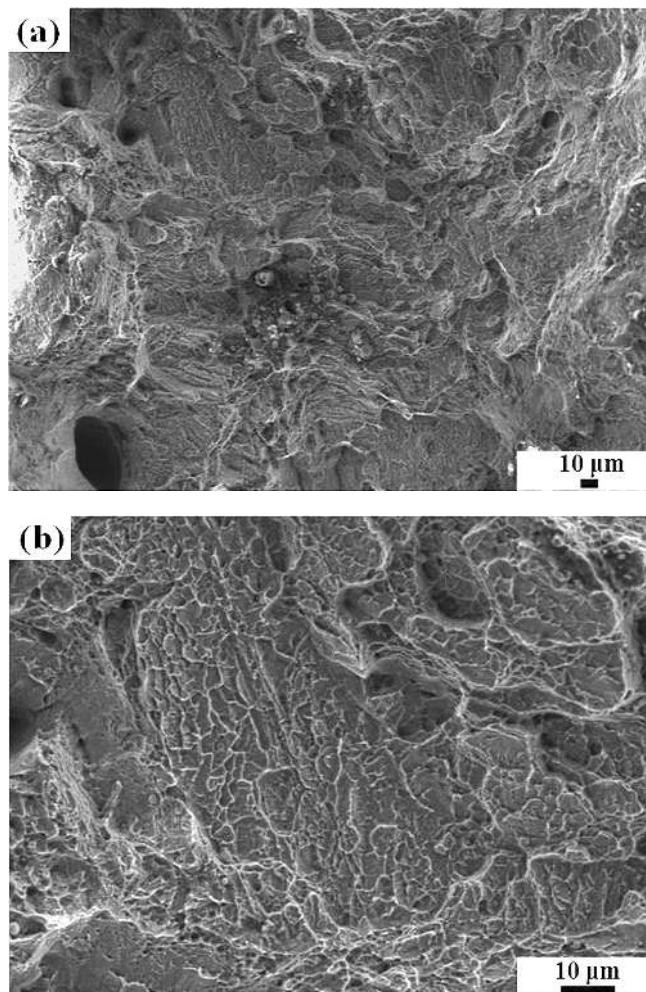


Fig. 6. Tensile fracture morphologies of the SLMed Ti-6Al-4V sample with an LED of $66.7 \text{ J} \cdot \text{mm}^{-3}$.

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